

4.2 JPEG

JPEG is a format that is standard for storing images across a wide range of formats and applications. It is used as a base for more specific applications such as the MPEG video format. Digital cameras typically store images in the JPEG / EXIF format. The exif data of an image store information such as the camera make and model, aperture, shutter speed, flash status used and other details of the photograph. Most digital cameras use this format as an alternative to their native raw file format.

The major advantage with the JPEG format is that it is a lossy image format. This means that depending on the compression, different amount of detail can be stored in the image. The more the compression, the lesser will be the details that are manifested on the image. This also means that saving a file over and over again will result in a proportional amount of depreciation in the quality of the image.

Although compressing the image by compromising on quality is possible on a computer, cameras do not offer this option. Different resolutions are offered in most cameras so that you can choose a suitable option between quality and image size. If you are going on a long trip without carrying too many memory cards, save the images in the camera in a medium or small size of the image. If, however, you are shooting at a location where it is easy to transfer images to a computer, you can afford to use the largest image size available.

The JPEG image is saved in the camera after processing the raw data from the image sensor. This means that once the sensor receives the raw information, the camera processes certain algorithms like the file dimension settings, the white balance, demosaicing, red-eye reduction, certain image parameters, and then saves the file on the camera. Essentially, every JPEG file that is stored in your camera is not of the highest possible quality your camera can produce, because of the loss of information that occurs during saving the image, something that is inherent to the format.

To avoid this, store images in the raw data format where available. Most cameras come with imaging software that can work with the raw data files in the cameras. Another major problem with JPEGs is the appearance of artefacts on com-